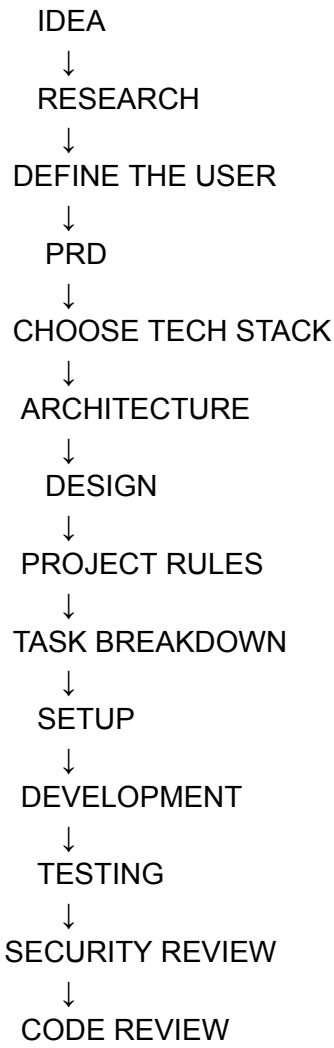
Vibe Coding: A Complete Beginner-to-Production Guide

From Idea → Planning → AI Coding → Testing → Deployment → Production

---

1. The Complete Vibe Coding Workflow

Before writing any code, understand the entire process.



↓
PREVIEW DEPLOYMENT
↓
QA TESTING
↓
PRODUCTION DEPLOY
↓
MONITORING
↓
ITERATION

Never skip directly from:

IDEA → AI → DEPLOY

---

2. Step 1: Define What You Want to Build

Before opening your AI coding tool, answer five questions.

1. What problem are you solving?

Example:

> College students struggle to organize assignments, notes and deadlines.

2. Who is the user?

Example:

> BCA, BSc CS and BTech students.

3. What is the main outcome?

Example:

> Give students one place to manage their academic work.

4. What is the MVP?

MVP means **Minimum Viable Product**.

For example:

Authentication
Dashboard
Notes
Assignments
Deadlines

Don't start with:

AI Tutor
Community
Payments
Mobile App
Gamification
Social Feed

Build the core product first.

5. What is NOT part of the first version?

This is equally important.

Example:

Out of scope:
- Mobile application
- Payments
- AI chatbot
- Social features

This prevents AI from continuously expanding the project.

---

3. Step 2: Research Before Coding

Don't ask AI to build an idea that you haven't researched.

Research:

Users

- Who needs this?
- What are their problems?
- What alternatives already exist?

Competitors

Look at:

- Features
- UI
- Pricing
- User experience
- Strengths
- Weaknesses

Technical feasibility

Check:

- APIs
- Authentication
- Database requirements
- Third-party services
- Hosting
- Cost

AI tools can help with research

You can use:

- ChatGPT
- Perplexity
- Google
- Official documentation
- GitHub
- Stack Overflow

For technical decisions, prefer **official documentation** over random blog posts.

---

4. Step 3: Choose Your AI Coding Tool

Don't use the same tool for every project.

For beginners

Replit

Good when you want:

- Browser-based development
- Minimal setup
- Quick prototypes
- Full-stack experimentation

Lovable

Good for:

- Web apps
- SaaS prototypes
- UI-heavy applications
- Rapid MVP development

Bolt

Good for:

- Fast web application prototypes
- JavaScript/TypeScript projects
- Browser-based development

Cursor

Good when you want:

- More control
- Existing repositories
- Larger projects
- Local development
- Professional development workflows

Claude Code

Good for:

- Terminal-based development
- Larger codebases
- Repository-level tasks
- Agentic development workflows

---

6. Recommended Beginner Stack

If you're serious about learning development through vibe coding, this is a good starting stack:

AI IDE
Cursor

Frontend
Next.js

Language
TypeScript

Styling
Tailwind CSS

Database
PostgreSQL / Supabase

Authentication
Supabase Auth

Version Control
Git + GitHub

Testing
Playwright

Deployment
Vercel

7. Step 4: Install the Development Environment

For a typical web project, install:

Required

1. Cursor or VS Code
2. Git
3. Node.js LTS
4. npm
5. GitHub account
6. Modern web browser

Optional

Docker
GitHub CLI
Postman / Bruno
Database client
Vercel CLI

Verify your installation:

```
node --version
npm --version
git --version
```

---

8. Step 5: Create Your Project

Create a project folder:

```
mkdir student-dashboard
cd student-dashboard
```

Initialize Git:

```
git init
```

Then create a GitHub repository.

From the beginning, your project should have version control.

Why?

Because AI can break your project.

Git gives you the ability to:

Change
↓
Test
↓
Break something
↓
Compare
↓
Rollback

---

9. Step 6: Create Your Project Documentation

This is one of the most important parts of structured vibe coding.

Before asking AI to build features, create your project context.

A professional project can have:

```
project/
│
├── docs/
│   ├── PRD.md
│   ├── ARCHITECTURE.md
│   ├── DESIGN.md
│   ├── TEST_PLAN.md
│   ├── SECURITY.md
│   ├── DECISIONS.md
│   └── MEMORY.md
│
├── .cursor/
│   └── rules/
│
├── src/
├── tests/
│
```

```
├── README.md
├── TASKS.md
├── .env.example
└── .gitignore
```

You don't need all of these for a tiny project.

For a beginner project, start with:

PRD.md
ARCHITECTURE.md
DESIGN.md
RULES.md
TASKS.md
README.md
.env.example

---

10. Step 7: Create `PRD.md`

What is PRD?

PRD = Product Requirements Document

It defines:

> **What are we building and why?**

Example:

```
# Product Requirements Document

## Product
StudentHub

## Problem
Students struggle to manage their academic
notes, assignments and deadlines.

## Target Users
BCA, BSc CS and BTech students.

## Goal
```

Create a centralized academic productivity dashboard.

Core Features

1. Authentication
2. Dashboard
3. Notes
4. Assignments
5. Deadlines

MVP

- Signup
- Login
- Dashboard
- Create notes
- Edit notes
- Delete notes
- Create assignments

Out of Scope

- Payments
- AI tutor
- Mobile app
- Social feed

Success Criteria

A user should be able to:

1. Create an account
2. Login
3. Create a note
4. Edit a note
5. Delete a note
6. Create an assignment
7. Mark an assignment complete

Remember:

PRD = WHAT + WHY

---

11. Step 8: Create `ARCHITECTURE.md`

This defines:

How will the application work?

Example:

```
# Architecture

## Frontend
Next.js + TypeScript

## Styling
Tailwind CSS

## Backend
Next.js server-side functionality

## Database
Supabase PostgreSQL

## Authentication
Supabase Auth

## Deployment
Vercel

## Architecture

User
↓
Next.js UI
↓
Server Action / API
↓
Supabase
↓
PostgreSQL
```

Define your folder structure too:

```
src/
├── app/
```

```
├── components/
├── features/
├── services/
├── lib/
├── types/
└── utils/
```

Also define architectural rules.

Example:

- UI components should not contain database logic.
- Database operations belong in services.
- Authentication must be verified server-side.
- Reusable UI should be placed in components.
- Business logic should remain separate from UI.

Remember:

Architecture = HOW

---

12. Step 9: Create `DESIGN.md`

This gives AI a consistent visual system.

Without it, you may get:

Page 1 → rounded cards

Page 2 → square cards

Page 3 → different buttons

Page 4 → completely different typography

Define:

Design System

Style
Modern
Minimal

Professional

Typography
Inter

Colors
Primary: #6366F1
Background: #F8FAFC
Text: #0F172A
Muted: #64748B

Buttons

Primary
Secondary
Destructive

Cards

Border radius: 12px

UX Requirements

- Mobile responsive
- Loading states
- Empty states
- Error states
- Accessible forms

Remember:

DESIGN.md = HOW IT SHOULD LOOK AND FEEL

---

13. Step 10: Create `RULES.md`

This is your project's **AI rulebook**.

Example:

Development Rules

General

- Use TypeScript.
- Reuse existing components.
- Do not duplicate logic.
- Keep functions small.
- Do not modify unrelated files.

Before Coding

- Read the relevant project documentation.
- Inspect existing implementation.
- Reuse existing functionality where possible.
- Make a plan for large changes.

UI

- Follow DESIGN.md.
- Maintain responsive design.
- Include loading states.
- Include error states.
- Include empty states.

Security

- Never expose API keys.
- Validate user input.
- Verify authorization server-side.

Testing

- Add tests for important functionality.
- Run tests after implementation.
- Fix failing tests before continuing.

Git

- Make small commits.
- Use descriptive commit messages.

For Cursor, these rules can also be organized inside:

.cursor/rules/

For example:

```
.cursor/
└── rules/
    ├── general.mdc
    ├── frontend.mdc
    ├── backend.mdc
    └── testing.mdc
```

---

14. Step 11: Create `TASKS.md`

Never ask AI to build the entire application in one prompt.

Break the project into tasks.

```
# Tasks

## Phase 1: Setup

- [ ] Initialize project
- [ ] Configure TypeScript
- [ ] Configure Tailwind
- [ ] Configure Git

## Phase 2: Authentication

- [ ] Create signup page
- [ ] Create login page
- [ ] Configure authentication
- [ ] Protect dashboard
- [ ] Test authentication

## Phase 3: Notes

- [ ] Create database table
- [ ] Create notes service
- [ ] Create notes UI
- [ ] Create note form
- [ ] Add edit functionality
- [ ] Add delete functionality
- [ ] Add tests
```

Then work like this:

TASK-001
↓
Implement
↓
Test
↓
Review
↓
Mark complete
↓
TASK-002

---

15. Step 12: Create `DECISIONS.md`

This stores important technical decisions.

Example:

```
# Architecture Decisions

## ADR-001

Decision:
Use Supabase for the database.

Reason:
Provides PostgreSQL, authentication and backend
services without managing our own infrastructure.

## ADR-002

Decision:
Use Next.js instead of React + Express.

Reason:
The application does not require a separate backend
and we want a unified full-stack application.
```

This prevents AI from randomly changing architectural decisions later.

---

16. Step 13: Create `MEMORY.md`

Think of this as the project's current state.

Example:

```
# Project Memory

## Current Status

Authentication completed.

Notes feature in progress.

## Completed

- Project setup
- Git setup
- Database setup
- Authentication

## Current Task

TASK-012

## Known Issues

- Mobile navbar needs improvement.
- Notes loading state is incomplete.

## Next Step

Complete note creation.
```

A useful distinction is:

DECISIONS.md
Permanent decisions

MEMORY.md
Current project state

---

17. Step 14: Create `TEST_PLAN.md`

Define what "working" actually means.

```
# Test Plan

## Authentication

- User can signup
- User can login
- Invalid credentials show error
- Logged-out users cannot access dashboard

## Notes

- User can create note
- User can edit note
- User can delete note
- User cannot access another user's note

## Responsive

Test at:

- 375px
- 768px
- 1440px

This becomes your testing checklist later.
```

---

18. Step 15: Create `SECURITY.md`

For production projects:

```
# Security Requirements

## Authentication

Private routes require authentication.
```

Authorization

Users can only access resources they own.

Secrets

Never expose secrets to client-side code.

Database

Use appropriate access policies.

Input

Validate all user input.

APIs

Validate request body and parameters.

File Uploads

Validate:

- File type
- File size
- Filename

Security should not be added five minutes before deployment.

---

19. Step 16: Create `.env.example`

Never put real API keys inside your source code.

Instead:

```
DATABASE_URL=
SUPABASE_URL=
SUPABASE_ANON_KEY=
OPENAI_API_KEY=
STRIPE_SECRET_KEY=
```

Then your local environment contains the real values:

.env.local

Your repository should contain:

.env.example

not your actual secrets.

---

20. Step 17: Give the AI Project Context

Now open your AI coding tool.

Don't immediately say:

> Build my app.

Start with:

Read the following files before making any changes:

PRD.md
ARCHITECTURE.md
DESIGN.md
RULES.md
TASKS.md

Do not modify anything yet.

First:
1. Understand the product.
2. Understand the architecture.
3. Review the design system.
4. Review the development rules.
5. Review the current tasks.
6. Identify missing information.
7. Explain the implementation plan for TASK-001.

Do not write code yet.

This lets the AI understand the project before touching it.

---

21. Step 18: Build One Feature at a Time

Use this workflow:

Understand
↓
Plan
↓
Implement
↓
Test
↓
Review
↓
Commit

For example:

TASK-001
Create signup page

Then:

TASK-002
Implement signup logic

Then:

TASK-003
Add validation

Then:

TASK-004
Add authentication tests

---

22. Step 19: Use Vertical Slices

Instead of building:

Entire frontend
↓
Entire backend
↓
Database

Build a complete user flow.

Example:

Signup
↓
Login
↓
Dashboard
↓
Create Note
↓
Save Note
↓
Display Note

Then:

Edit Note
↓
Save
↓
Display updated note

Then:

Delete Note
↓
Confirm
↓
Database update
↓
UI update

This makes debugging much easier.

---

23. Step 20: Use a Structured Prompt

A good coding prompt has six parts:

CONTEXT
TASK
FILES
CONSTRAINTS
ACCEPTANCE CRITERIA
TESTING

Example:

CONTEXT

We are building a student notes application.
Read PRD.md, ARCHITECTURE.md and RULES.md.

TASK

Implement note creation.

FILES

Relevant files:
src/features/notes/
src/services/
src/types/

CONSTRAINTS

- Follow the existing architecture.
- Reuse existing components.
- Do not create a second database layer.
- Do not modify unrelated files.
- Validate input.

ACCEPTANCE CRITERIA

- Logged-in user can create a note.
- Title is required.
- Content is required.
- Invalid input shows an error.
- Note is saved to the database.

- New note appears in the UI.

TESTING

Add appropriate tests.
Run lint.
Run type checking.
Run relevant tests.

After implementation, report:

1. Files changed
2. What was implemented
3. Tests executed
4. Remaining issues

---

24. Step 21: Never Give AI Huge Tasks

Bad

Build the entire SaaS application.

Better

Build authentication.

Best

Create the signup UI.

Do not implement authentication yet.

Follow DESIGN.md.

Requirements:
- Email field
- Password field
- Confirm password field
- Validation
- Loading state
- Error state
- Mobile responsive

Do not modify unrelated files.

Small tasks give AI less room to misunderstand your project.

---

25. Step 22: Test Every Feature

After implementing a feature:

Code
↓
Lint
↓
Type check
↓
Unit tests
↓
Integration tests
↓
E2E tests

Type check

npm run typecheck

Lint

npm run lint

Tests

npm test

Build

npm run build

Your exact commands depend on the project's configuration.

---

26. Step 23: Use End-to-End Testing

E2E testing checks the application from the user's perspective.

For example:

Open website
↓
Signup
↓
Login
↓
Create note
↓
Refresh
↓
Verify note exists
↓
Edit note
↓
Delete note
↓
Logout

Tools such as Playwright are useful for automating browser-based tests.

---

27. Step 24: Review AI-Generated Code

Don't assume:

> "It compiled, so it's correct."

Ask the AI:

Review the implementation against:

PRD.md
ARCHITECTURE.md
DESIGN.md
RULES.md
TEST_PLAN.md
SECURITY.md

Check:

- Correctness
- Architecture
- Security
- Error handling
- Accessibility
- Responsive design
- Performance
- Code duplication

Do not modify anything yet.

Report all issues first.

Then fix them one by one.

---

28. Step 25: Learn How to Debug With AI

When something breaks, don't say:

> Fix this.

Give the AI structured information.

ERROR

[paste error]

EXPECTED BEHAVIOR

The user should be redirected to the dashboard.

ACTUAL BEHAVIOR

The page shows a 500 error.

STEPS TO REPRODUCE

1. Login
2. Click Dashboard
3. Error appears

CONSTRAINT

Do not change the database schema.

Then ask:

Do not modify code yet.

Find the root cause.

Explain:
1. What is failing?
2. Why is it failing?
3. Which file is responsible?
4. What is the smallest fix?
5. How will we test the fix?

Then:

Implement the smallest fix.

Do not refactor unrelated code.

Run the relevant tests.

---

29. Step 26: Use Git Throughout Development

After a working feature:

git status

Then:

git add .

Then:

git commit -m "feat: add note creation"

Then:

git push

Use branches for larger projects:

main
│
├── feature/auth
├── feature/notes
├── feature/search
└── fix/mobile-navbar

This protects your working code from AI-generated mistakes.

---

30. Step 27: Prepare for Deployment

Before deployment, verify:

Functionality

[ ] Signup
[ ] Login
[ ] Logout
[ ] CRUD operations
[ ] Forms
[ ] Error handling
[ ] Loading states
[ ] Empty states

UI

[ ] Mobile
[ ] Tablet
[ ] Desktop
[ ] Accessibility
[ ] Keyboard navigation

Security

[ ] No secrets in Git
[ ] Authentication verified
[ ] Authorization verified
[ ] Database security configured
[ ] Input validation

[ ] API validation

Code

[ ] TypeScript passes
[ ] Lint passes
[ ] Tests pass
[ ] Production build passes

---

31. Step 28: Deploy to Preview First

Your deployment pipeline should look like:

LOCAL
↓
PREVIEW
↓
QA
↓
PRODUCTION

Never make production your first testing environment.

For example:

feature/notes
↓
Preview deployment
↓
Test
↓
Fix
↓
Merge
↓
Production

---

32. Step 29: Deploy to Production

For a Next.js application, Vercel is one common deployment option.

The general workflow is:

GitHub repository
↓
Connect to Vercel
↓
Configure environment variables
↓
Deploy
↓
Preview URL
↓
Test
↓
Production

You can also use the Vercel CLI when appropriate.

vercel

After testing:

vercel --prod

---

33. Step 30: Configure Environment Variables

Your local environment might contain:

.env.local

Production should have its own values.

For example:

Development
DATABASE_URL=development-db

Preview
DATABASE_URL=staging-db

Production

DATABASE_URL=production-db

Don't assume your local environment is identical to production.

---

34. Step 31: Production QA

After deployment, test the actual live application.

Don't only test localhost.

Check:

Live URL
↓
Signup
↓
Login
↓
Core features
↓
Database operations
↓
Error handling
↓
Mobile
↓
Desktop

Also test:

- Refreshing pages
- Direct URLs
- Logged-out access
- Invalid inputs
- Slow network
- Empty database
- Wrong credentials

---

35. Step 32: Monitor the Application

Deployment isn't the end.

The production lifecycle is:

Deploy
↓
Monitor
↓
Collect feedback
↓
Find bugs
↓
Fix
↓
Test
↓
Deploy again

Consider adding:

- Error tracking
- Analytics
- Performance monitoring
- Database backups
- Uptime monitoring

---

36. Step 33: Maintain Project Documentation

As the project changes, update:

TASKS.md
MEMORY.md
DECISIONS.md
README.md
ARCHITECTURE.md

Don't let documentation describe an application that no longer exists.

The AI needs accurate context.

---

37. The Recommended Project Structure

For a serious beginner project:

```
student-dashboard/
│
├── docs/
│   │
│   ├── PRD.md
│   ├── ARCHITECTURE.md
│   ├── DESIGN.md
│   ├── TEST_PLAN.md
│   ├── SECURITY.md
│   ├── DECISIONS.md
│   └── MEMORY.md
│
├── .cursor/
│   └── rules/
│       ├── general.mdc
│       ├── frontend.mdc
│       ├── backend.mdc
│       └── testing.mdc
│
├── src/
│   ├── app/
│   ├── components/
│   ├── features/
│   ├── services/
│   ├── lib/
│   ├── types/
│   └── utils/
│
├── tests/
│   ├── unit/
│   ├── integration/
│   └── e2e/
│
├── public/
│
```

```
├── .env.example
├── .gitignore
├── README.md
├── TASKS.md
├── package.json
└── ...
```

---

38. What Each File Does

File	Purpose	Stage
`PRD.md`	What are we building?	Planning
`ARCHITECTURE.md`	How will we build it?	Planning
`DESIGN.md`	How should it look?	Planning
`RULES.md`	How should AI code?	Planning
`TASKS.md`	What should we build next?	Development
`DECISIONS.md`	Why did we make this decision?	Development
`MEMORY.md`	What is the current project state?	Development
`TEST_PLAN.md`	How do we verify it?	Testing
`SECURITY.md`	How do we protect it?	Development
`.env.example`	What configuration is required?	Setup
`README.md`	How do humans use the project?	Documentation

---

39. The Minimum Setup for Beginners

If all of this feels overwhelming, start with only:

my-project/

```
│
├── PRD.md
├── RULES.md
├── TASKS.md
├── README.md
├── .env.example
│
└── src/
```

Then add:

ARCHITECTURE.md
DESIGN.md
TEST_PLAN.md
SECURITY.md
DECISIONS.md
MEMORY.md

as your project becomes more complex.

---

40. The Professional Vibe Coding Loop

For every feature, follow this exact loop:

```
1. READ
   ↓
2. UNDERSTAND
   ↓
3. PLAN
   ↓
4. IMPLEMENT
   ↓
5. TEST
   ↓
6. REVIEW
   ↓
7. FIX
   ↓
8. COMMIT
   ↓
9. UPDATE DOCUMENTATION
```

Then move to the next feature.
